

ROB Internal Logic

Overview

The Reorder Buffer (ROB) absorbs out-of-order read completions from MC Core and emits them in-order per AXID to Merge Logic. Depth is tied to `MC_RD_BUF = 64`. Since `RESP_READY` must always be held high, the ROB can never stall incoming completions — backpressure is instead applied by halting new read requests when occupancy reaches the watermark threshold.

Inputs: Tag Allocator (`alloc_req`, `AXID`, `seqnum`, `tag`); MC Core (`RESP_VALID`, `tag[7:0]`, `data[511:0]`, `status[3:0]`)

Outputs: Merge Logic (`data[511:0]`, `status[3:0]`, `AXID`, `last_beat`); Packet FIFO (`PKT_READY`); MC Core (`RESP_READY` tied high)

Blocks

Free List

- 64-bit bitmask. 1 = free, 0 = occupied.
- Allocate: priority encoder selects lowest free bit, clears it, outputs `free_slot_idx[5:0]`.
- Free: Emit Logic sets the bit at `slot_idx` via `free_req`.
- Outputs `free_slot_idx` to Lookup Table (write), ROB BRAM (init), and Head Pointer Array (first allocation).

Occupancy Counter

- Up/down counter, range 0–64.
- Incremented by `alloc_req` from Free List.
- Decrement by `free_req` from Emit Logic.
- Asserts `watermark_hit` when count $\geq \text{MC_RD_BUF} - \text{ROB_WATERMARK} = 38$.

Lookup Table

- 256-entry register file, addressed by `{AXID[5:0], seqnum[1:0]}` (8-bit address).
- Each entry stores `slot_idx[5:0]`.
- Write port: Tag Allocator writes `slot_idx` at `{AXID, seqnum}` on allocation.
- Read port: Completion Input reads `slot_idx` using `{tag[7:2], tag[1:0]}`.
- Clear: Emit Logic clears entry at `{AXID, seqnum}` after emit.

ROB BRAM

- Dual-port BRAM, 64×517 bits per slot: `valid[0]`, `data[511:0]`, `status[3:0]`.

- Write port: Completion Input writes `data`, `status`, `valid=1` at `slot_idx`.
- Write port (init): Free List writes `valid=0` at `free_slot_idx` on allocation.
- Read port: Head Pointer Array supplies `head[AXID]` as read address; all 64 head slots checked in parallel each cycle.
- 1-cycle read latency — Emit Logic must account for this in its valid check timing.

Completion Input

- Receives `RESP_VALID`, `tag[7:0]`, `data[511:0]`, `status[3:0]` from MC Core.
- Extracts `AXID = tag[7:2]`, `seqnum = tag[1:0]`.
- Reads `slot_idx` from Lookup Table using `{AXID, seqnum}`.
- Writes `data`, `status`, `valid=1` to ROB BRAM at `slot_idx`.

Head Pointer Array

- 64 registers of 6 bits each, one per AXID. Stores slot index of next expected beat.
- Written on first allocation: `head[AXID] = free_slot_idx`.
- Advanced by Emit Logic after each emit: incremented to next expected slot.
- All 64 head pointers drive the ROB BRAM read port simultaneously each cycle.

Emit Logic

- Reads `valid` bits from all 64 head slots out of ROB BRAM each cycle.
- Priority encoder selects lowest AXID whose head slot has `valid=1`.
- On selection: reads `data` and `status` from BRAM; emits to Merge Logic.
- After emit:
 - Clears `valid` in BRAM slot.
 - Advances `head[AXID]` in Head Pointer Array.
 - Sends `free_req + slot_idx` to Free List (sets bitmask bit).
 - Decrements Occupancy Counter via `free_req`.
 - Clears Lookup Table entry at `{AXID, seqnum}`.
- Maximum one emission per cycle (single R channel path).

PKT_READY Control

- Purely combinational.
- Asserts `PKT_READY=0` to Packet FIFO when `watermark_hit` is high.
- Halts new read requests to MC Core; prevents ROB overflow since `RESP_READY` cannot be deasserted.

Allocation Path

1. Tag Allocator sends `alloc_req` (pulse) with `AXID`, `seqnum`, `tag`.
2. Free List pops `free_slot_idx` (priority encoder on bitmask), clears the bit.
3. Free List writes `slot_idx` to Lookup Table at `{AXID, seqnum}`.
4. Free List initializes ROB BRAM slot: `valid=0` at `free_slot_idx`.
5. Free List stores `free_slot_idx` into `head[AXID]` (first allocation for that AXID only).
6. Occupancy Counter increments.

Completion Path

1. MC Core asserts `RESP_VALID` with `tag[7:0]`, `data[511:0]`, `status[3:0]`.
2. Completion Input extracts `AXID = tag[7:2]`, `seqnum = tag[1:0]`.
3. Completion Input reads `slot_idx` from Lookup Table.
4. Completion Input writes `data`, `status`, `valid=1` to ROB BRAM at `slot_idx`.
5. `RESP_READY` is tied high — MC Core never stalled on this path.

Emit Path

1. Each cycle, all 64 `head[AXID]` values drive the ROB BRAM read port.
2. BRAM returns `valid` bits for all 64 head slots (1-cycle latency).
3. Priority encoder in Emit Logic selects lowest `AXID` with `valid=1`.
4. Emit Logic reads `data` and `status` from selected slot; forwards to Merge Logic with `AXID` and `last_beat`.
5. Slot is freed: `valid` cleared, `head[AXID]` advanced, Free List bit set, Occupancy Counter decremented, Lookup Table entry cleared.

Backpressure & Overflow Prevention

Signal	Description
<code>watermark_hit</code>	Asserted when occupancy ≥ 38 . Drives <code>PKT_READY</code> control.
<code>PKT_READY=0</code>	Stops Packet FIFO from issuing new read requests to MC Core.
<code>RESP_READY=1</code>	Tied high permanently. ROB can never stall MC Core completions.

Watermark headroom = `ROB_WATERMARK` = 26. Ensures all requests already accepted by MC Core can return completions without overflowing the ROB.

Key Parameters

Parameter	Value
ROB depth	64 entries (<code>MC_RD_BUF</code>)
Slot width	517 bits (1 valid + 512 data + 4 status)
Free list width	64 bits
Lookup table	256 entries $\times$ 6-bit slot index
Head pointers	64 $\times$ 6-bit registers
Watermark	38 (64 – 26)
Max emissions	1 per cycle

Timing Notes

- ROB BRAM has 1-cycle read latency. Emit Logic valid check must be pipelined accordingly — register the read address, use valid bit one cycle after presenting `head[AXID]`.
- Priority encoder across 64 valid bits is the critical path in Emit Logic. May need pipelining at high frequencies.

- Allocation and completion can occur in the same cycle — write port (completion) and init write (allocation) are independent BRAM ports.
- `watermark_hit` is combinational from the Occupancy Counter — no additional latency before `PKT_READY` deasserts.